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Introduction

Getting started with Pattern Recognition (and Computational Learning)

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Agenda

1 The Pattern Recognition Process

2 Learning Paradigms

3 Initial definitions and general notation

4 What's next?

The Pattern Recognition Process

PR

- Research area that deals with classification and description of objects
- Multidisciplinary nature, interfaces with:
  - Statistics
  - Engineering
  - Artificial Intelligence
  - Computer Science
  - Data Mining
  - Signal and Image Processing
  - etc...

- Applications:
  - Classical problems — automatic character recognition, medical diagnostics, monitoring bank clients
  - Current topics such as recommendation systems, facial recognition, etc.

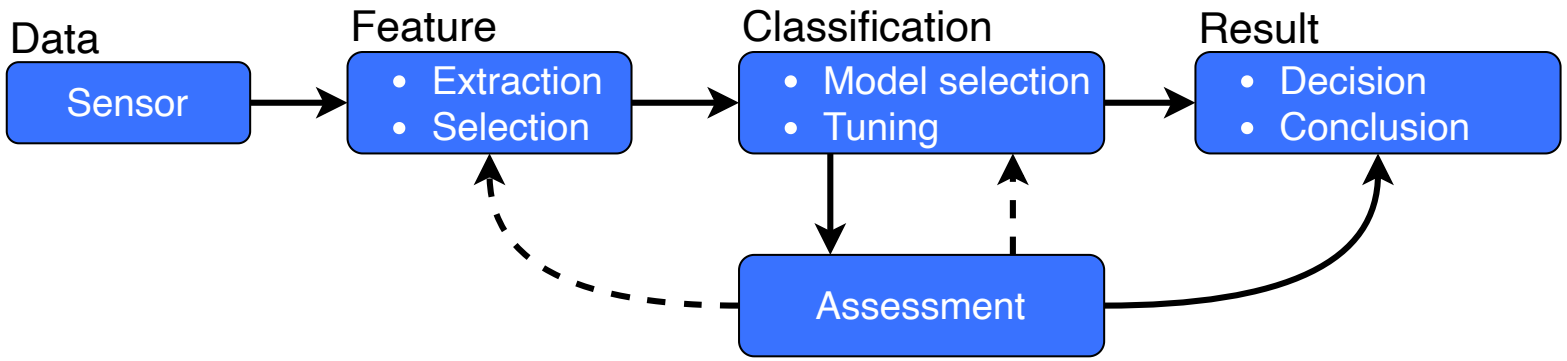
PR *vs.* ML

What's the difference?

- Machine Learning (ML) has become a popular research area with intense discussions and developments
- A common question is: *What is the relationship between Pattern Recognition and Machine Learning?*
- Although opinions may differ, we can say that:
  - Both aim at constructing association rules and making decisions based on data
  - PR emphasizes the "complete process"
  - ML is more focused on the computational methods
- It is important to highlight that both areas are strongly related!

Keep in mind!

- A research investigation in PR does not consist solely of applying methods to a given dataset, which will magically reveal previously hidden information
- A *PR process* encompasses several stages: problem formulation — data collection and representation — classification — assessment — interpretation of results



- A pattern is a vector whose components represent values for specific attributes of the object.
- The attributes are previously observed/acquired through sensors.
- We want to associate each pattern to a specific class (or group), which carries a semantic meaning.
- The process of associating a pattern to a class is called classification, which typically requires a model or rule to perform such a task.
- As intermediate steps, we highlight:
  - Attribute processing — "feature extraction" & "feature selection"
  - Classification evaluation — done persistently and systematically

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Learning Paradigms

- The core of PR is modeling a rule that classifies patterns into groups or classes
- Learning Paradigm — refers to how the modeling of such rules is carried out
- Among several types, the paradigms supervised, unsupervised, and semi-supervised are commonly used in PR
- Other paradigms include reinforcement learning, instance-based, and evolutionary learning — less common

- Supervised and unsupervised learning differ in how classification rules are modeled
  - Supervised — modeling is based on available information
  - Unsupervised — modeling is based on analogies among patterns
- The available information is a set of "labeled" patterns (known class)
- When no class is associated with a pattern, it is considered "unlabeled"
- A set of labeled patterns is called a *training set*

- If the problem classes are predefined, supervised learning is preferable
- However, various constraints may arise, making it impractical to obtain a labeled dataset...
- In such cases, unsupervised learning naturally emerges as an alternative
- It is important to note that the analogies inferred in this case limit classification to producing subgroups composed of similar patterns, without assigning labels to them
- This limitation was one of the starting points for the development of the semi-supervised paradigm
- ...a "middle way" between supervised and unsupervised learning, as it incorporates both labeled and unlabeled data in the modeling process

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Initial Definitions and General Notation (pt.1)

Scalars are denoted by italic lowercase letters

Vectors are denoted by bold lowercase letters

Matrices are denoted by bold uppercase letters

Sets are denoted by calligraphic uppercase letters

Pattern is denoted by the vector $\mathbf{x} = [x_1, \ldots, x_n]$, where each component expresses a specific characteristic of the pattern

Attribute corresponds to a specific feature of a given pattern; in this sense, x_i, with $i = 1, \ldots, n$, corresponds to the i-th attribute of $\mathbf{x}$

Feature space refers to the set containing the attribute vectors — that is, the patterns — and is denoted by $\mathcal{X}$

Initial Definitions and General Notation (pt.2)

Class is a label assigned to patterns to qualify them in relation to a specific property — classes are denoted by ω

Class space refers to the set $\Omega = \{\omega_1, \omega_2, \ldots, \omega_c\}$, whose elements are the classes

Class indicator is an index used to indirectly associate a pattern with a class — this indicator is usually denoted by $y \in \mathcal{Y} \subset \mathbb{N}^*$; if a pattern $\mathbf{x}$ is associated with indicator y, and $y = k$, then $\mathbf{x}$ should be associated with class ω_k

Indicator space is the set of discrete values $\mathcal{Y} = \{1, \ldots, c\}$, whose elements serve as indicators of the classes in $\Omega = \{\omega_1, \ldots, \omega_c\}$

Initial Definitions and General Notation (pt.3)

Association generically expressed by the tuple $(\mathbf{x}, \omega)$, meaning the pattern $\mathbf{x}$ has a property that classifies/labels it into class ω. The notation $(\mathbf{x}, y)$ is also valid and indicates that $\mathbf{x}$ is labeled as ω_k whenever $y = k$

Training set refers to the set $\mathcal{D} = \{(\mathbf{x}_i, y_i) \in \mathcal{X} \times \mathcal{Y} : i = 1, \ldots, m\}$, whose elements are patterns with known class labels. As per the mathematical definition of a set, no duplicate elements are allowed, so each tuple in $\mathcal{D}$ is distinct by index i, not by the values of $\mathbf{x}_i$ or y_i

Initial Definitions and General Notation (pt.4)

Classification set represented by $\mathcal{I}$, where $\mathcal{I} \subset \mathcal{X}$, and is composed of patterns to be classified

Classifier is usually represented as a function $f\colon \mathcal{X} \to \mathcal{Z}$; for supervised learning, f is modeled using the training set $\mathcal{D}$ and $\mathcal{Z} \equiv \Omega$. For unsupervised learning, f is modeled from the data in $\mathcal{I}$, yielding a partition of this set into different subsets $\mathcal{K}_1, \ldots, \mathcal{K}_k$ such that $\bigcup_{j=1}^{k} \mathcal{K}_j = \mathcal{I}$ and $\bigcap_{j=1}^{k} \mathcal{K}_j = \emptyset$

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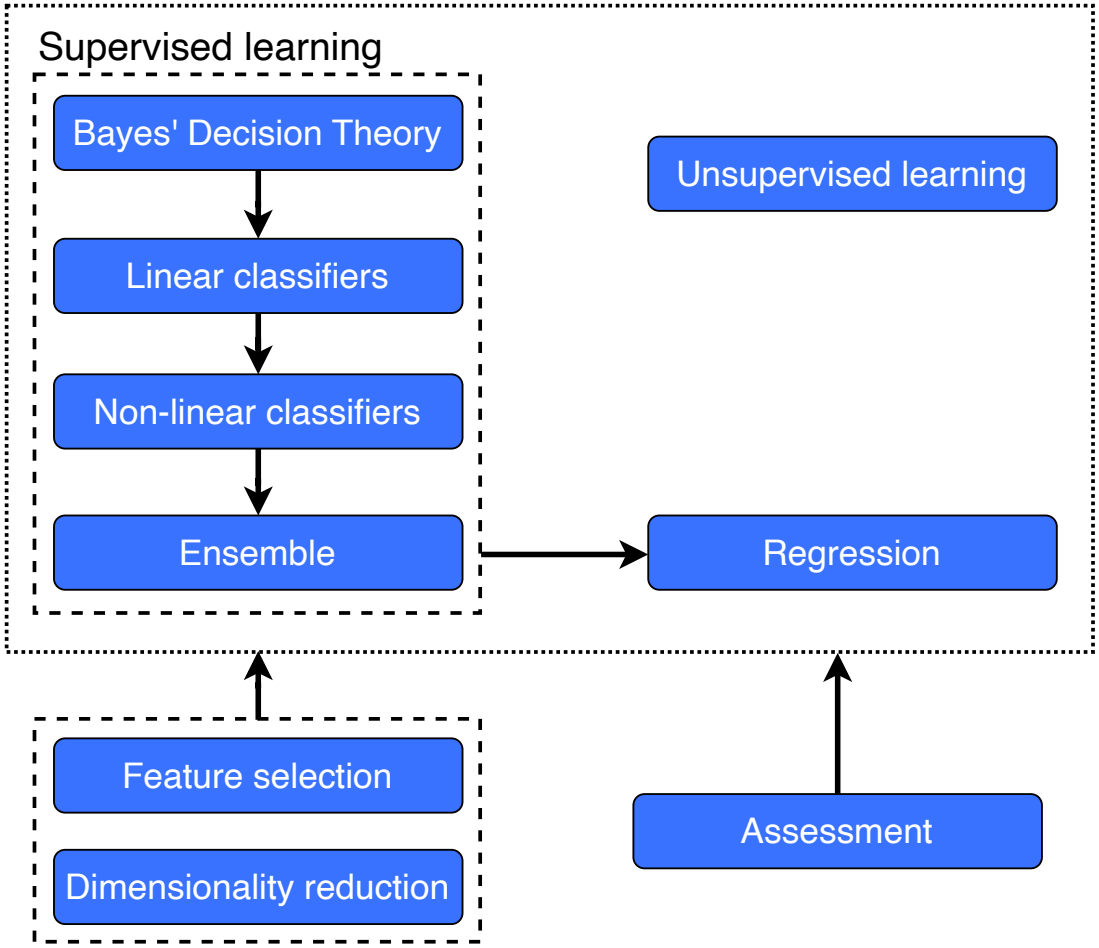
Supervised learning
Bayes' Decision Theory
Linear classifiers
Non-linear classifiers
Ensemble
Unsupervised learning
Regression
Feature selection
Dimensionality reduction
Assessment

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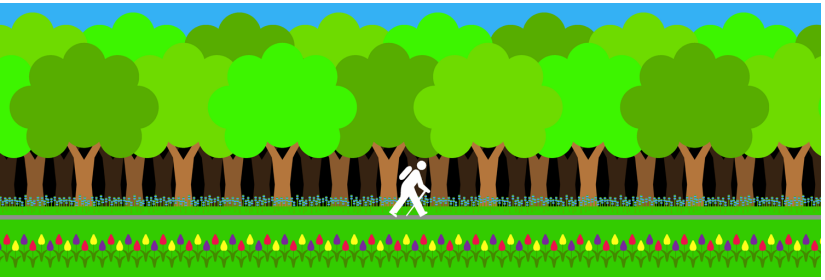
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